Do Now · retrieval

Lesson 096 · Little Man Computer: Instructions and the Accumulator

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the ACC for?

Q2

What does STA X do?

Q3

What does INP do?

Q4

Half adder A1,B1: what Sum and Carry?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

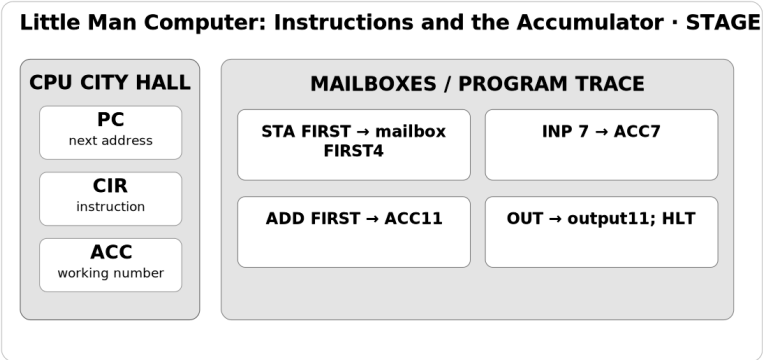
Core idea 2

Apply — INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

Core idea 3

Evaluate — OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LMC — Little Man Computer, simplified stored-program model.

mailbox — Memory location.

ACC — Accumulator for working results.

STA — Store ACC into memory.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

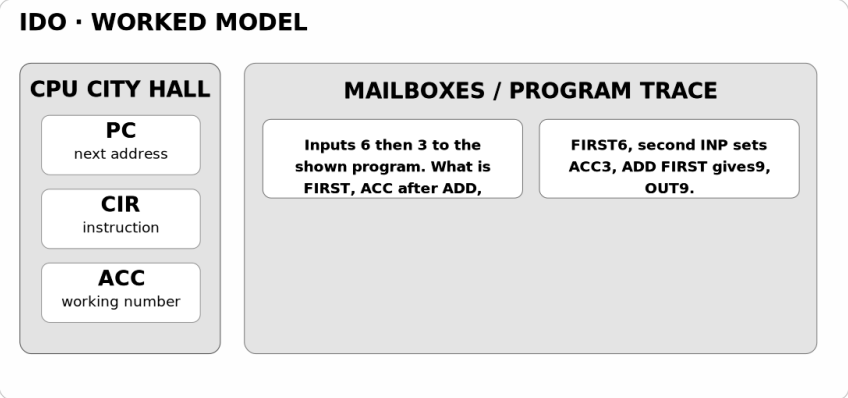
Problem

Inputs 6 then 3 to the shown program. What is FIRST, ACC after ADD, and OUT?

Follow the worked method

1. Follow the reasoning — FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
After first input store FIRST=4; second input ACC9; SUB FIRST → ACC5; OUT5.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace program above for inputs 4,7: ACC after each INP, STA and ADD; output. [6]

Question 2

Write an LMC program that reads one integer, stores it as N, then loads and outputs N. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare LDA FIRST, STA FIRST and ADD FIRST in relation to ACC and mailbox; give distinct effects. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

STA FIRST overwrites ACC with the mailbox's value.

Use key vocabulary from memory

1. Explain LMC in your own words.

2. Explain mailbox in your own words.

3. Explain ACC in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.